

CO5 ASSESSMENT 1

CONSTRAINT-BASED PROBLEM SOLVING

Answers with Sample Images for Each Question

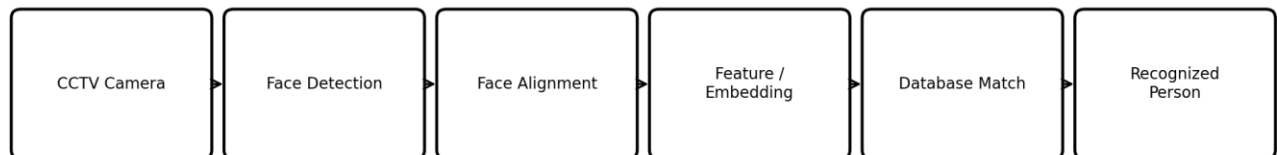
1. Real-Time Face Recognition for Campus Security

Design an OpenCV pipeline that captures live CCTV frames, detects faces, extracts facial features, compares them with registered identities, and displays the final identification.

- Image acquisition: Capture frames continuously using OpenCV VideoCapture.
- Preprocessing: Resize frames for speed, normalize illumination when necessary, and apply mild denoising.
- Face detection: Use an OpenCV Haar cascade or a lightweight DNN face detector to locate faces.
- Feature extraction: Align the detected face and generate a facial feature representation/embedding.
- Recognition: Compare the live feature vector with stored registered-person features using a distance or similarity measure. Use a conservative threshold and an Unknown class to reduce false recognition.
- Temporal confirmation: Confirm an identity across multiple frames before accepting it.
- Optimization: Detect at reduced resolution, track faces between detection frames, keep the registered gallery in memory, and use lightweight models/hardware acceleration where available.

This pipeline addresses live capture, lighting variation, feature extraction, recognition, low-cost computation, and false-recognition reduction.

Sample Image - Face Recognition



Example: a camera frame is processed until the registered person's identity is displayed.

Sample illustrative image for Question 1

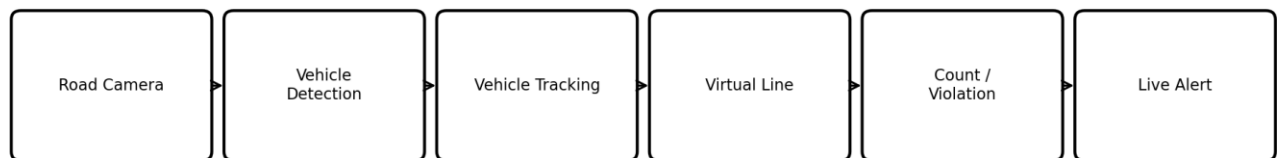
2. Intelligent Traffic Violation Detection System

Design an OpenCV video-analysis pipeline to detect, track, count, and analyze vehicles in a roadside camera stream.

- Video acquisition: Read live frames using OpenCV VideoCapture.
- Preprocessing: Resize frames and apply mild Gaussian/median filtering to reduce camera noise.
- Vehicle detection: For a fixed camera, background subtraction such as MOG2 can identify moving foreground objects; an object detector can be used when classification is required.
- Motion analysis: Threshold the foreground mask and remove small noise.
- Morphology and contours: Opening removes isolated noise and closing fills small gaps; contours provide vehicle regions.
- Tracking: Assign a persistent ID and match each vehicle across consecutive frames using centroid/bounding-box information.
- Line crossing: Draw a virtual line and count a vehicle once when its tracked centroid crosses it.
- Abnormal movement: Estimate direction from centroid changes and trigger an alert for wrong-way or restricted-region movement.
- Visualization: Display bounding boxes, IDs, line, count, and alerts.

Shadows, dynamic backgrounds, camera noise, and multiple vehicles are handled through filtering, adaptive background modeling, area thresholds, persistent tracking IDs, and one-time line-crossing logic.

Sample Image - Traffic Detection



Example: tracked vehicles cross a predefined line and the system increments the count.

Sample illustrative image for Question 2

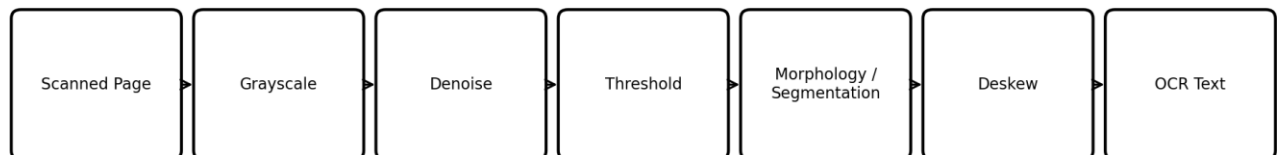
3. Automated Document Processing System

Design an OpenCV preprocessing pipeline that converts poor-quality scanned pages into clean images suitable for OCR.

- Grayscale conversion: Reduce the page to one intensity channel for simpler processing.
- Filtering: Use median or Gaussian filtering to reduce scanner noise while preserving characters.
- Illumination correction: Reduce uneven background brightness so text has more uniform contrast.
- Thresholding: Use global thresholding for uniform pages or adaptive thresholding for uneven illumination.
- Morphological operations: Opening removes small unwanted components; closing can repair small gaps in text.
- Segmentation: Detect connected components/contours and group them into text lines, words, columns, or regions.
- Geometric correction: Estimate skew or perspective and apply rotation/perspective transformation to straighten the page.
- OCR: Pass the cleaned and segmented regions to an OCR engine and store the recognized text.
- Batch processing: Process documents independently and use parallel workers where appropriate.

Each stage improves OCR by making characters cleaner, separating text from background, and correcting the page geometry before recognition.

Sample Image - Document Processing



Example: a noisy scanned page is cleaned and converted into an OCR-ready document.

Sample illustrative image for Question 3

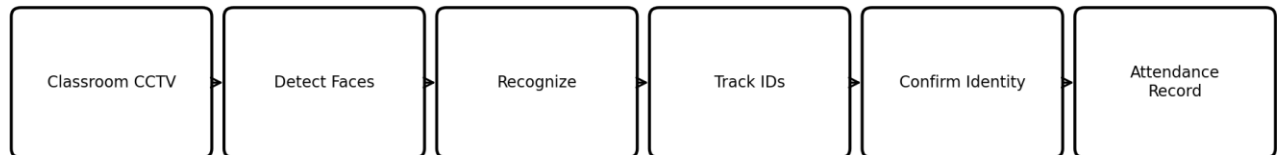
4. Real-Time Multi-Person Tracking and Attendance System

Design a system that detects multiple faces, recognizes registered students, tracks identities through video, and prevents duplicate attendance.

- Capture classroom CCTV frames continuously.
- Detect all visible faces in each frame.
- Align and normalize each face, then extract a feature embedding.
- Compare the embedding with registered student embeddings using a threshold; uncertain matches remain unknown.
- Assign a tracking ID to each person and associate the recognized identity with that ID.
- Continue the track during short face disappearances or head movement using tracking information.
- Require several consistent recognition results before confirming identity.
- Maintain an attendance set/database for the current class session. If a student's ID is already present, do not create another attendance record.
- Use detection less frequently and tracking between detections to reduce processing delay.

This separation of detection, recognition, tracking, and attendance confirmation prevents repeated marking and improves identity stability.

Sample Image - Multi-Person Attendance



Example: each student receives a tracking ID and attendance is recorded only once.

Sample illustrative image for Question 4

5. Smart Home Surveillance and Intrusion Detection

Design an OpenCV surveillance pipeline for a fixed entrance camera that distinguishes meaningful movement from noise and background changes.

- Capture the fixed-camera stream.
- Build an adaptive background model using a method such as MOG2 and generate a foreground mask.
- Filter sensor noise with Gaussian/median filtering.
- Apply thresholding and morphological opening/closing to clean the foreground mask.
- Extract contours and discard very small regions using an area threshold.
- Track persistent moving objects across frames using centroids or bounding boxes.
- Define an entrance/restricted region of interest (ROI).
- Generate an intrusion alert only when a sufficiently large and persistent tracked object enters the ROI.
- Use persistence, object size, direction, and dwell time to distinguish suspicious activity from brief disturbances.

Shadows can be reduced through foreground/shadow handling and shape/area checks. Illumination changes are handled with adaptive background modeling. Trees and other dynamic background objects are reduced using area thresholds, persistence checks, and ROI restriction. Camera shake can be treated as a scene-wide disturbance rather than an intrusion.

Sample Image - Home Intrusion Detection



Example: a person enters the protected entrance region and an intrusion alert is generated.

Sample illustrative image for Question 5

Conclusion

The five OpenCV designs use preprocessing, detection, feature or motion analysis, tracking, decision logic, and visualization according to the requirements in the assessment. The figures are sample pipeline illustrations intended to make each solution easier to understand.